Machine Learning-based Smart Helmet for Personalized Construction Monitoring

Progress Presentation - 02



Project ID: R26-SE-042

Research Aim:

- Personalized AI-driven worker safety monitoring framework for construction environments.
  - **Combines environmental and physiological monitoring** into a single integrated risk assessment system.
  - Uses **worker-specific physiological baseline analysis** instead of fixed threshold values.
  - **Machine learning-based classification** of worker states as Safe, Warning, or Critical.
  - Real-time alerts, adaptive monitoring, and AI-powered safety recommendations.



Construction Workers

- Receive real-time safety monitoring and emergency alerts.
- Get early warnings and safety recommendations during hazardous conditions.



Safety Officers / Supervisors

- Monitor worker conditions in real time through a centralized dashboard.
- Improve incident response and reduce reliance on manual supervision.

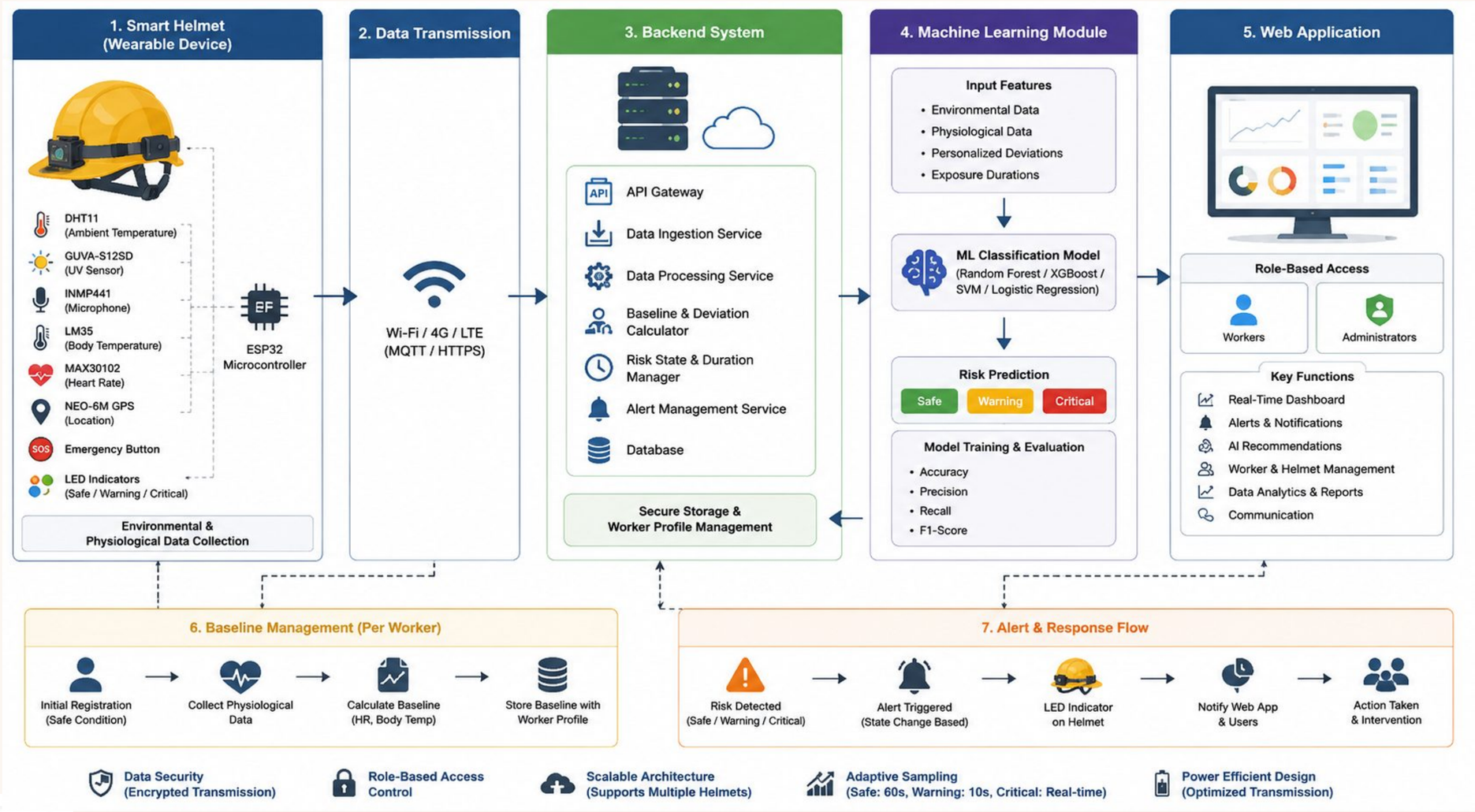
Existing Solutions & Their Limitations

System / Study	Environmental Monitoring	Physiological Monitoring	Machine Learning	Real-time Alerts	Personalization	Integrated Risk Analysis
IoT Smart Helmet (2025)	✔	✖	✖	✔	✖	✖
BLE-LoRa Smart Helmet	✔	✖	✖	✔	✖	✖
Smart Helmet and Jacket System	✔	✔	✖	✔	✖	✖
Coal Miner Wearable Device	✔	✔	✖	✔	✖	✖
SafeHelmet-AI	✔	✔	✔	✔	✖	Partial
AI Deep Learning Helmet	✔	✔	✔	✔	✖	✖
Smart Wear Guardian	✔	✔	✔	✔	✖	✖
Personalized Wearable Sensors Study	✔	✔	✔	✖	✔	✖
Thermal Comfort Wearable Study	✔	✔	✔	✖	✔	✖
Proposed System	✔	✔	✔	✔	✔	✔

Key Limitations

- Reliance on fixed threshold-based alerts
- Lack of personalization for individual workers
- Environmental and physiological data analyzed separately
- Limited real-time adaptability
- Mostly prototype-level (limited real-world deployment)

Solution Architecture - “ProSafe” Smart Helmet



Impact of “ProSafe”

Early Detection

- ProSafe enables early identification of safety risks by continuously monitoring both environmental conditions and worker physiological responses.

Personalized Safety

- Worker-specific physiological baselines provide more personalized monitoring and reduce dependence on generic fixed thresholds.

Integrated Analysis

- Integrated analysis of environmental and physiological data addresses the current research gap of fragmented safety monitoring.

Faster Response

- Machine-learning-based Safe, Warning, and Critical classifications help supervisors recognize changing risk conditions before they develop into more serious incidents.

Better Decisions

- Real-time alerts, worker location, historical data, and centralized monitoring provide safety officers with better information for faster and more informed decisions.

Technologies | Ethical Consideration | Standards

Technologies

Smart Helmet:

- ESP32 + integrated sensors

Backend

- Node.js + Express 5
- MongoDB + Mongoose

Machine Learning

- Python + Flask
- XGBoost / Scikit-learn

Web Application

- React 18 + Vite
- Recharts + Leaflet

Real-time & External Services

- Server-Sent Events
- Open-Meteo API

Ethical Consideration

- Privacy of physiological and location data
- Role-based access to sensitive worker information
- Informed consent when collecting worker-related research data
- Human oversight of ML predictions and safety recommendations

Standards

- **Secure authentication:** JWT + bcrypt password hashing
- **Authorization**: Role-Based Access Control
- **API security:** CORS and protected endpoints
- **Software testing:** Automated frontend and backend tests
- **Data validation:** Helmet-side and backend-side validation
- **Version control & modular architecture:** maintainable separation of frontend, backend, ML and firmware

AI Usage

Limitations and Chalenges

1. **Battery Capacity:** Continuous sensing, Wi-Fi communication, and GPS operation increase power consumption and limit operating time.
2. **Sensor Reliability in Construction Environments :** Dust, vibration, movement, heat, moisture, and physical impacts can affect sensor readings or damage components.
3. **Network Dependency :** Real-time communication with the backend depends on network availability, which may be inconsistent at some construction sites.
4. **Physiological Sensor Accuracy:** Heart-rate and body-temperature readings can be affected by worker movement, sensor positioning, and contact quality.
5. **Model Generalization:** The current ML model is developed and evaluated using the available research dataset and may not represent every worker, site condition, or construction environment.

Commercialization

Target Market

Primary

- Construction companies and contractors
- Infrastructure development firms

Future Expansion

- Mining
- Manufacturing
- Other industrial safety environments

Revenue Model

- **Smart Helmet Sale:** One-time hardware purchase
- **Dashboard Subscription:** Monthly / annual SaaS subscription for monitoring, analytics and alerts
- **Maintenance & Support:** Optional maintenance, updates and support services

Work Breakdown

Proposal Presentation

- ✅ Research Planning & Requirement Analysis

PP1

- ✅ Wearable Smart Helmet Development
- ✅ Data Collection & Preprocessing
- ✅ Machine Learning Model Development
- ½ Backend System Development

PP2

- ✅ Backend System Development
- ✅ Web Application Development
- ✅ Integration and Testing

Final Presentation

- Impact Evaluation - Interviewing
- Documentation

DEMONSTRATION

Multi-Modal Safety Intelligence Component

- IT Number: IT21200892
- Name: Kariyapperuma M.V.



(01) “ProSafe” Device Implementation

Sensors Used

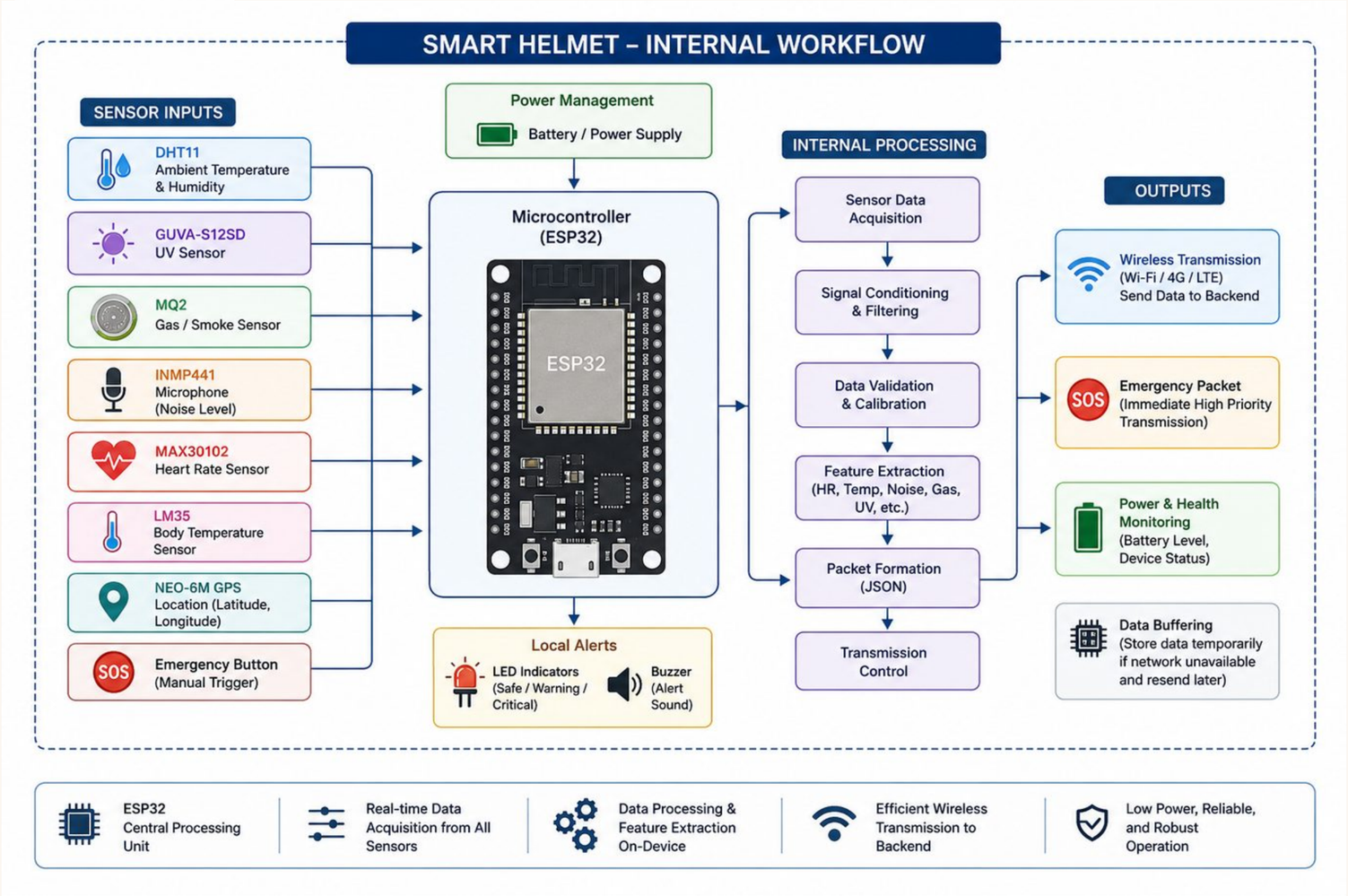
Ambient Temperature	:	DHT11
UV exposure	:	GUVA-S12SD UV detection sensor module
Gas/ VOC	:	MQ2 Gas sensor
Noise	:	INMP441 MEMS microphone
Body Temperature	:	LM35
Pulse Rate	:	MAX30102
Location	:	Neo-6M GPS module
Processing and Control Unit	:	Node MCU ESP32 microcontroller
Battery	:	Lithium-ion battery unit

Steps followed:

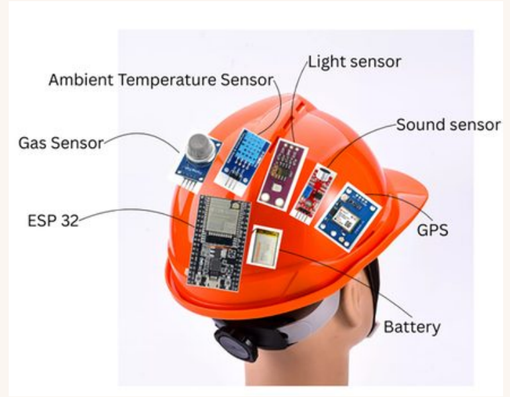
1. Hardware Integration
2. Firmware & Communication Development
3. Sensor Calibration & Testing
4. Real-Time Monitoring Implementation

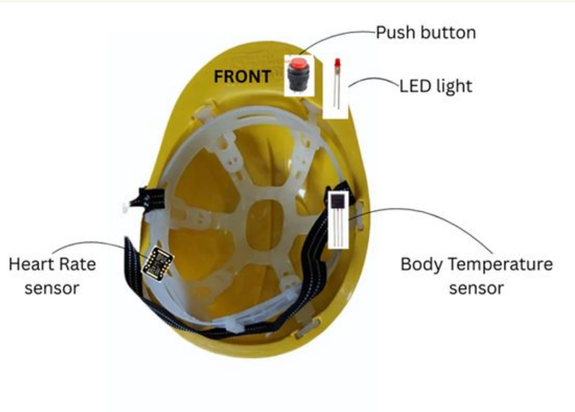
- Normal Packet sending interval: 60s
- Emergency packet sending: Immediately

01 “ProSafe” Device Implementation



Helmet Design



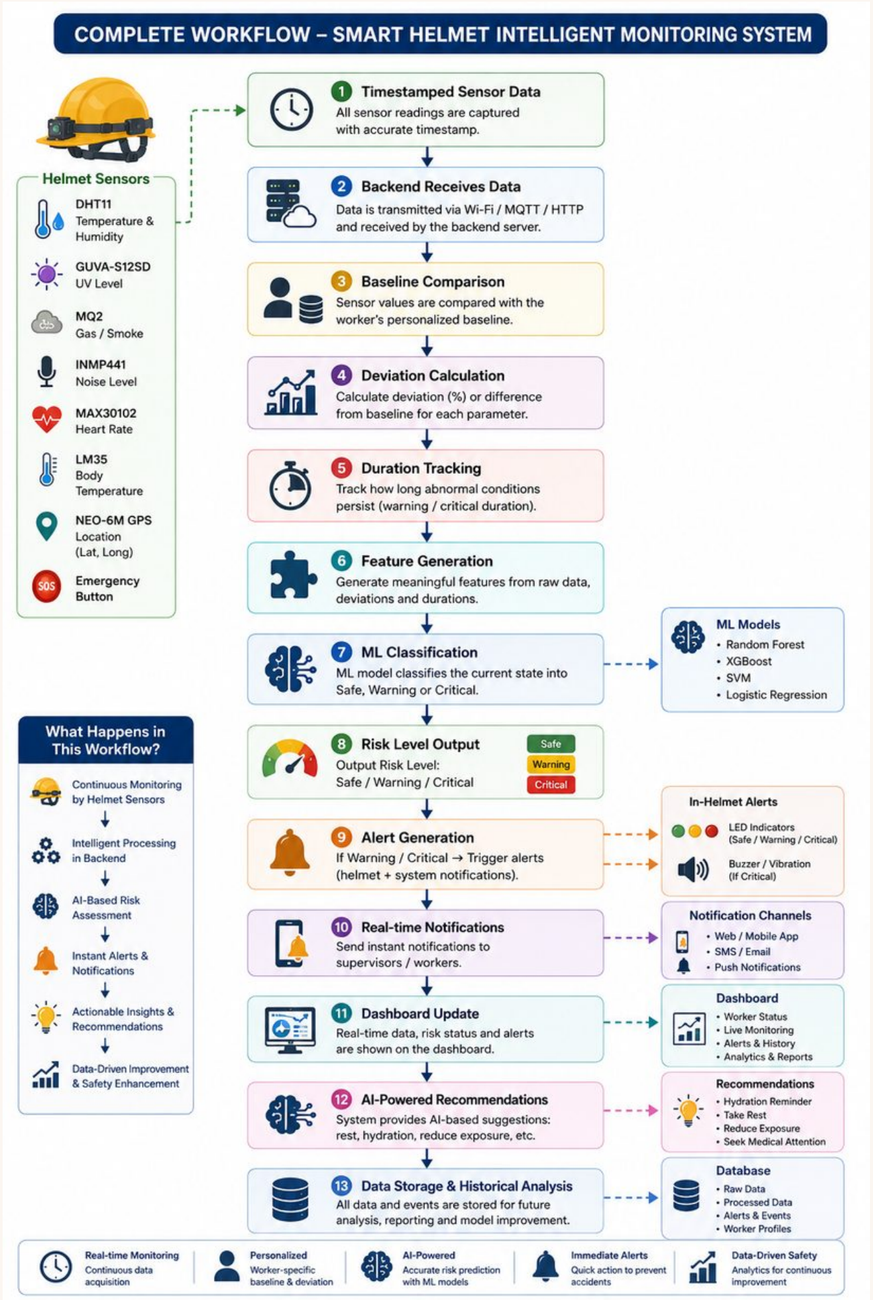


Threshold Values for Environmental Data

- Environmental conditions are evaluated using predefined occupational safety thresholds.

Baseline Comparison

- Each worker has a personalized physiological baseline created during the initial registration phase.
- Current heart rate and body temperature values are compared against the worker's normal baseline values.
- Enables worker-specific safety monitoring instead of fixed physiological thresholds.



Deviation Calculation

- Calculates how much the current physiological readings differ from the personalized baseline.
- Deviation percentage is used to support the classification of Safe, Warning, and Critical states.

[(Current - Baseline)/Baseline]*100

Duration Tracking

- Tracks how long abnormal conditions persist continuously.
- Prevents false alarms caused by temporary fluctuations or physical activity spikes.

Ambient Temperature				
	Safe	**Warning**	**Critical**	**Standard**
Threshold Value	<27 °C	27 - 35°C	> 35 °C	OSHA - NIOSH
Action	• LED – green	• LED – yellow • Waning alert	• LED – red • Critical alert	

Gas Sensor				
	Safe	**Warning**	**Critical**	**Standard**
Threshold Value	<150 ppm	150-300 ppm	>300 ppm	NIOSH fire/smoke early warning guidance
Action	• LED – green	• LED – yellow • Warning alert	• LED – red • Critical alert	

Sound Sensor				
	Safe	**Warning**	**Critical**	**Standard**
Threshold Value	<80 dB	80-85 dB	85-90dB	OSHA PEL
Action	• LED – green	• LED – yellow • Waning alert	• LED – red • Critical alert	

Light Sensor				
	Safe	**Warning**	**Critical**	**Standard**
Threshold Value	<3 UV index	3-8 UV index	>8 UV index	WHO
Action	• LED – green	• LED – yellow • Warning alert	• LED – red • Critical alert	

(02) Machine Learning Model Implementation

Dataset - model 1, User 1, User 2

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	worker_id	timestamp	ambient_temp_c	uv_index	gas_ppm	noise_db	body_temp_c	heart_rate_bpm	baseline_hr_bpm	baseline_body_temp_c	hr_deviation_pct	body_temp_deviation_pct	risk_level
2	W046	1/6/2025 8:00	18.84	1.41	142.39	54.53	37.14	63.2	68	37.14	-7.06	0	Safe
3	W049	1/6/2025 8:00	19.47	0.97	81.33	57.98	35.79	76.3	78.1	36.28	-2.3	-1.35	Safe
4	W060	1/6/2025 8:01	39.71	2.46	112.87	72.53	37.08	85	87.3	36.62	-2.63	1.26	Critical
5	W090	1/6/2025 8:01	19.34	1.22	34.77	53.97	34.79	76.6	88.4	35.41	-13.35	-1.75	Safe
6	W060	1/6/2025 8:02	20.94	2.03	130.74	66.09	35.93	79.4	87.3	36.62	-9.05	-1.88	Safe
7	W048	1/6/2025 8:02	27.39	2.31	84.17	66.81	37.25	99.6	84.3	36.48	18.15	2.11	Warning
8	W071	1/6/2025 8:03	23.3	2.53	104.68	55.94	35.02	122.7	89.1	35.76	37.71	-2.07	Critical
9	W071	1/6/2025 8:03	22.81	2.34	65.9	73.25	35.75	121.5	89.1	35.76	36.36	-0.03	Critical
10	W001	1/6/2025 8:04	22.46	2.37	128.64	66.85	32.96	80.3	74.4	37.85	7.93	-12.92	Critical

+20,000 Records

Model Training; 80%
Model Testing: 20%

User 1 - 9000+ records

User 2 - 9000+ records

Risk Threshold

Safe State
- Physiological deviation below 15%,
- Environmental conditions within acceptable ranges.

Warning State
- Physiological deviation remains between 15%-30%,
- Prolonged environmental hazard exposure occurs.

Critical State
- Physiological deviation exceeds 30%,
- Severe environmental and physiological stress occur together.

Machine Learning Model Implementation

Why These Algorithms Were Chosen,

Random Forest

- Handles noisy sensor data effectively
- Learns complex relationships between environmental and physiological features
- Suitable for structured sensor datasets

XGBoost

- Provides high classification accuracy
- Efficient for tabular data and feature-based learning
- Handles combined risk patterns effectively

Support Vector Machine (SVM)

- Effective for multiclass classification problems
- Performs well with moderate-sized datasets
- Suitable for clear risk boundary identification

Logistic Regression

- Simple and computationally efficient baseline model
- Easy to interpret and compare with advanced models
- Useful for evaluating overall model improvements

Best Algorithm: XGBoost

```
============================================================
TRAINING & EVALUATING MODELS
============================================================

Training Random Forest...
  Accuracy  : 0.9755
  Precision : 0.9773
  Recall    : 0.9755
  F1-Score  : 0.9755

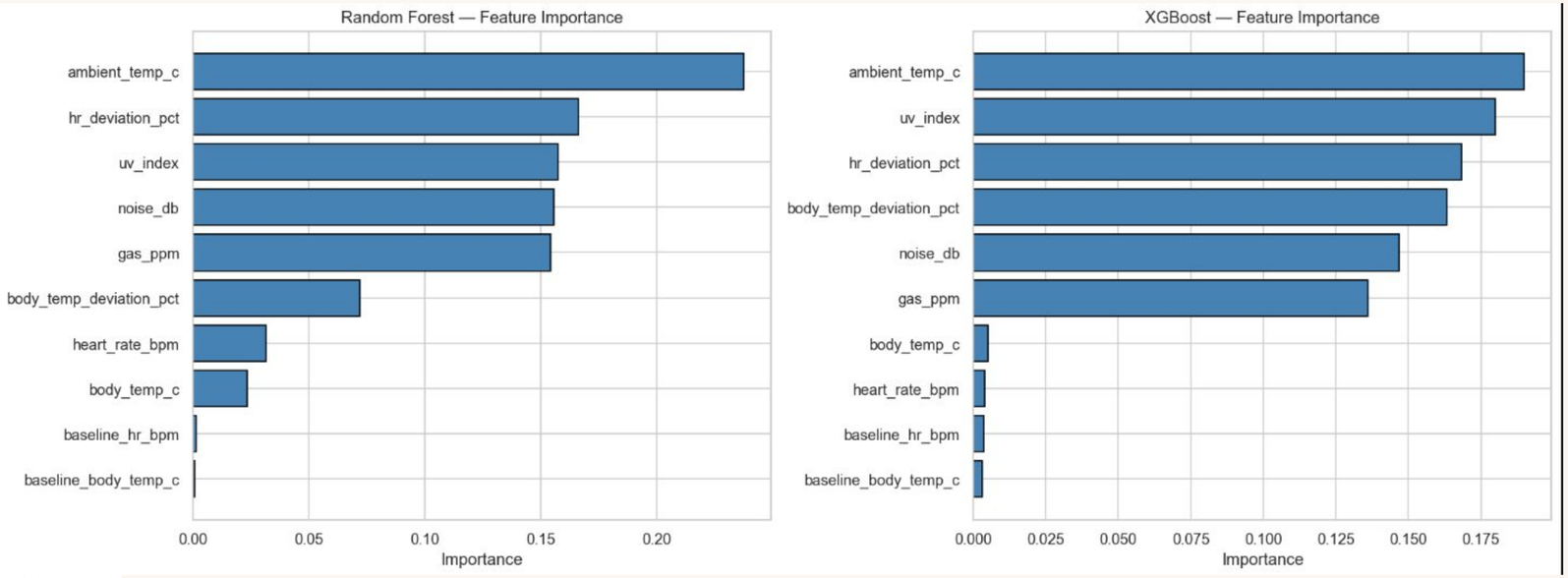
Training XGBoost...
  Accuracy  : 0.9890
  Precision : 0.9890
  Recall    : 0.9890
  F1-Score  : 0.9890

Training SVM...
  Accuracy  : 0.9255
  Precision : 0.9264
  Recall    : 0.9255
  F1-Score  : 0.9258

Training Logistic Regression...
  Accuracy  : 0.6408
  Precision : 0.6435
  Recall    : 0.6408
  F1-Score  : 0.6417
```

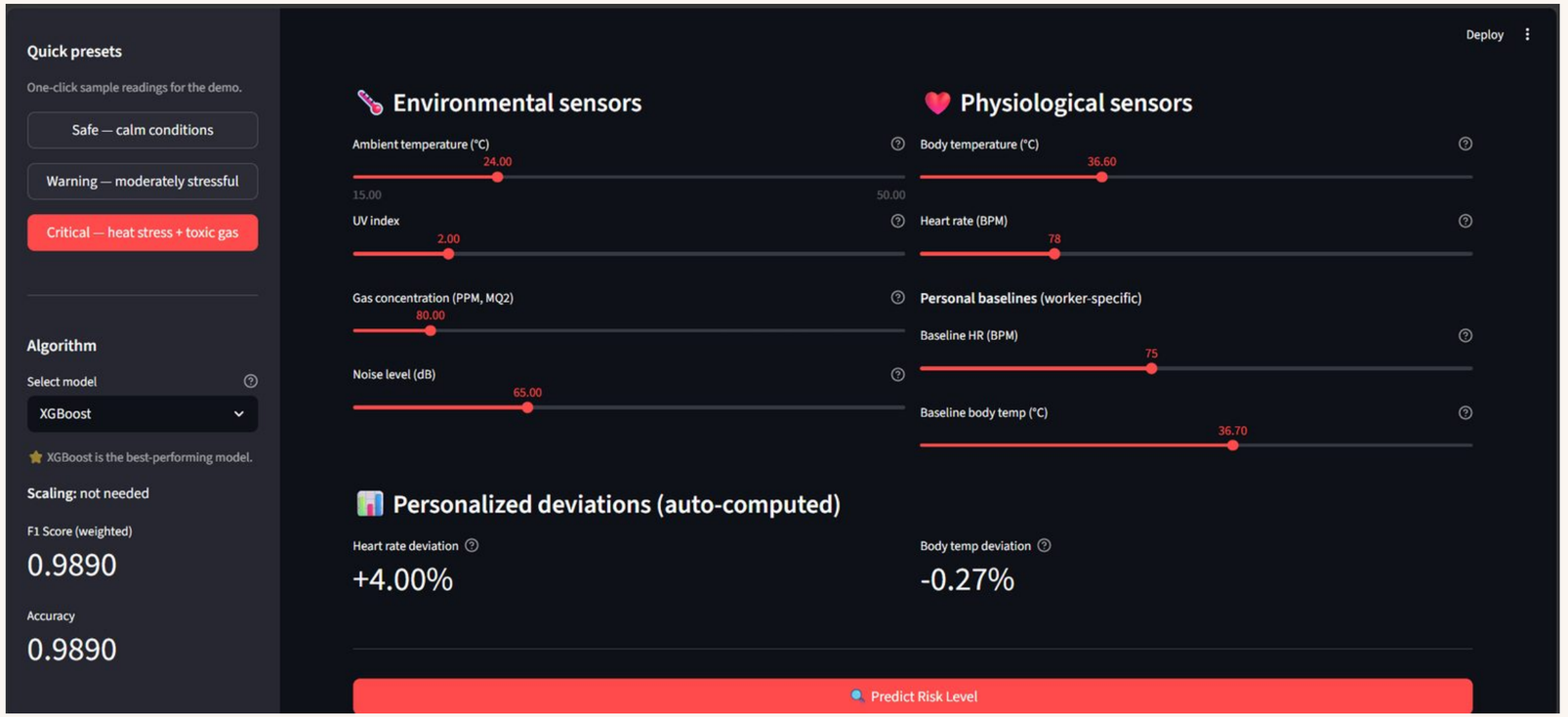
Machine Learning Model Implementation

```
5-FOLD STRATIFIED CROSS-VALIDATION
============================================================
  Random Forest          F1 = 0.9786 +/- 0.0043  (min=0.9753, max=0.9870)
  XGBoost                F1 = 0.9873 +/- 0.0014  (min=0.9857, max=0.9895)
  SVM                    F1 = 0.9262 +/- 0.0027  (min=0.9238, max=0.9315)
  Logistic Regression    F1 = 0.6372 +/- 0.0101  (min=0.6245, max=0.6513)
```



Feature Importance Distribution

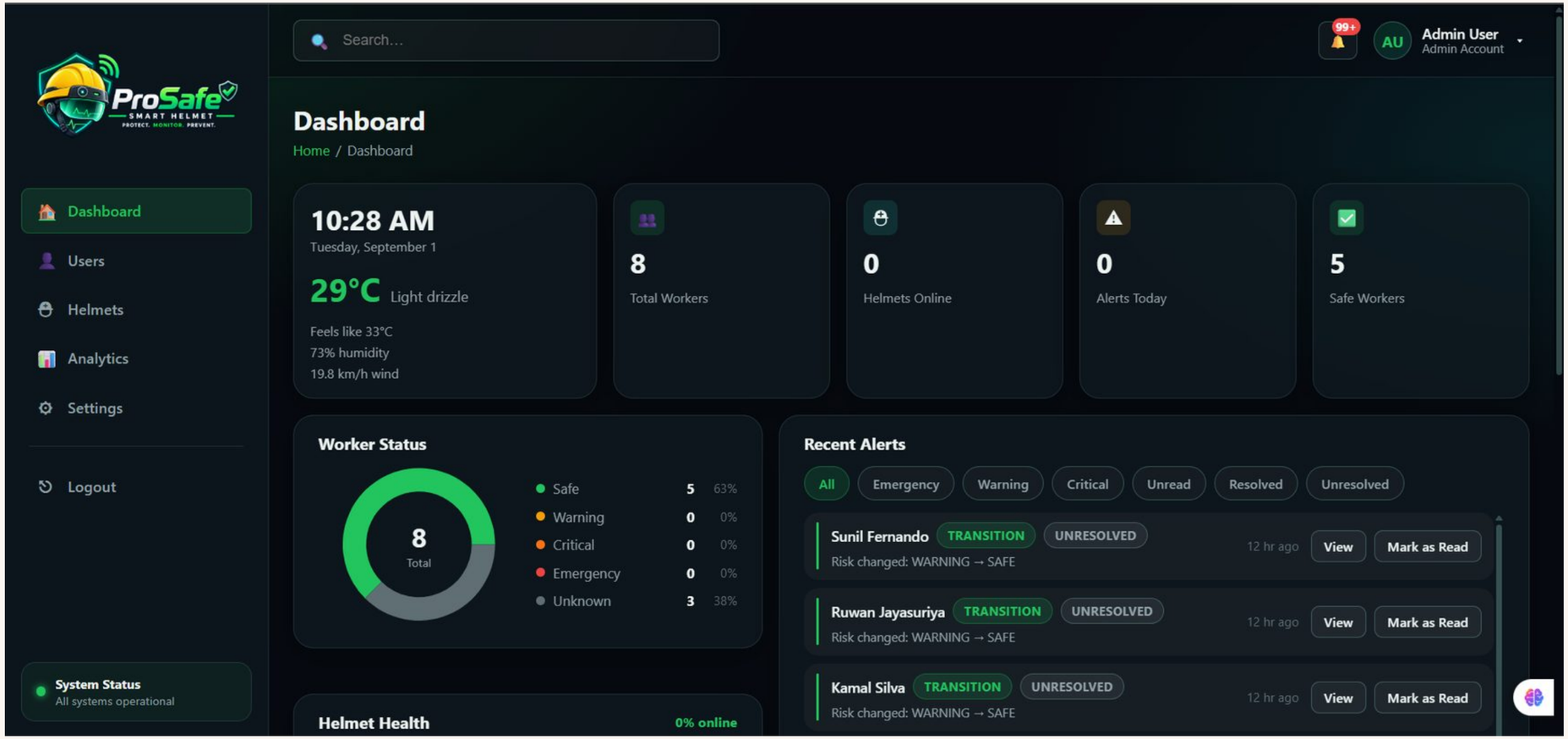
Machine Learning Model Implementation



Streamlit UI - For Demo

selected XG boost wrapped around a fask ml app

(03) Web Application Develpment

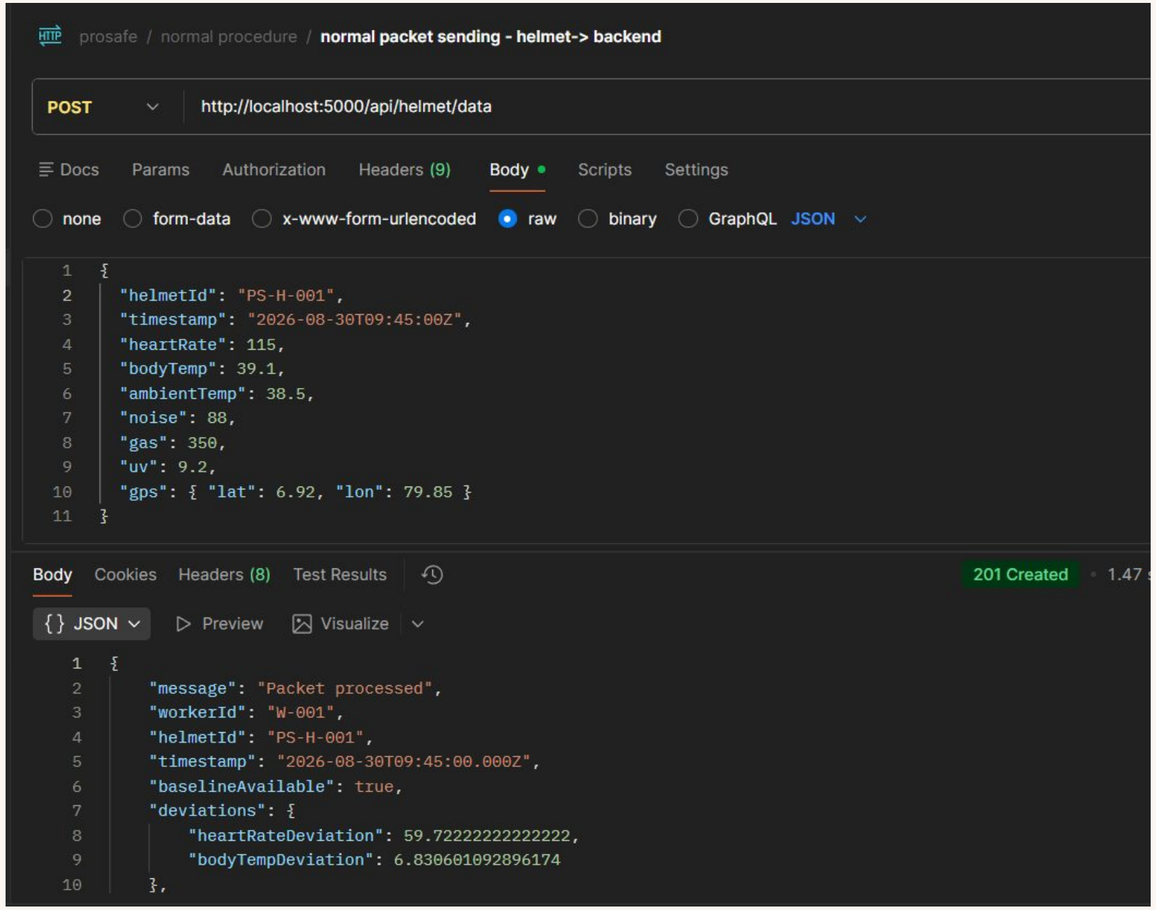


(04) Testing

```
✓ src/components/layout/NotificationBell.test.jsx (9) 1027ms
✓ src/components/routes/ProtectedRoute.test.jsx (2)
✓ src/components/routes/RoleRoute.test.jsx (3)
✓ src/components/users/CurrentConditionCard.test.jsx (6)
✓ src/components/users/DeleteUserModal.test.jsx (5) 498ms
✓ src/components/users/UserDetailView.test.jsx (14) 1835ms
✓ src/components/users/UserForm.test.jsx (5) 2822ms
✓ src/components/users/WorkerLocationCompactCard.test.jsx (5)
✓ src/pages/dashboard/AdminDashboard.test.jsx (7) 793ms
✓ src/pages/dashboard/WorkerDashboard.test.jsx (5) 866ms
✓ src/pages/analytics/AnalyticsPage.test.jsx (14) 3505ms
✓ src/pages/helmets/HelmetListPage.test.jsx (9) 6651ms
✓ src/components/users/sensors/EnvironmentalSensorModal.test.jsx (2) 362ms
✓ src/components/users/sensors/PersonalizedSensorModal.test.jsx (6) 688ms
✓ src/components/users/sensors/PredictionTimelineChart.test.jsx (2)
✓ src/components/users/sensors/SafetyPredictionModal.test.jsx (3) 397ms
✓ src/components/users/sensors/SensorHistoryChart.test.jsx (1)

Test Files  31 passed (31)
     Tests  194 passed (194)
  Start at  11:03:38
  Duration  14.19s (transform 8.60s, setup 12.51s, collect 38.26s, tests 33.12s, environment 78.52s, prepare 12.05s)
```

Frontend and backend are also supported by automated testing using **Jest/Supertest and Vitest/React Testing Library.**



API testing - Postman

Remaining work for PP2

- Impact Evaluation - Interviewing
- Documentation



Thank you

